

Lab 2

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We solved for an analytic solution for order of convergence. To do this, begin with the expression

$$\frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lambda. \quad (1)$$

Nikole taught us a cool trick. Let ε_k denote $|p_k - p^*|$, where p_k is the k^{th} iteration of an algorithm with p^* being the value we are searching for. Then:

$$\frac{\varepsilon_{k+1}}{\varepsilon_k^\alpha} \approx \lambda$$

$$\log\left(\frac{\varepsilon_{k+1}}{\varepsilon_k^\alpha}\right) \approx \log(\lambda)$$

$$\log(\varepsilon_{k+1}) - \log(\varepsilon_k^\alpha) = \log(\lambda)$$

$$\log(\varepsilon_{k+1}) - \alpha \log(\varepsilon_k) = \log(\lambda)$$

Now, consider:

$$\log(\varepsilon_{k+1}) - \alpha \log(\varepsilon_k) = \log(\lambda) \quad (2)$$

$$\log(\varepsilon_{k+1}) - \alpha \log(\varepsilon_k) = \log(\lambda). \quad (3)$$

We subtract 3 from 2 to obtain

$$\log(\varepsilon_{k+1}) - \log(\varepsilon_k) + \alpha \log(\varepsilon_{k-1}) - \alpha \log(\varepsilon_k) = 0.$$

Notice we have eliminated the asymptotic error constant λ . Hence, we obtain

$$\alpha = \frac{\log\left(\frac{\varepsilon_{k+1}}{\varepsilon_k}\right)}{\frac{\varepsilon_k}{\varepsilon_{k-1}}}. \quad (4)$$

For the later exercises, we turn the reader towards the git repository by *chuanweiheng-cell* entitled APPM4600. Thank you!